

REFLECTION PAPER

Advisor - Prof. Vlad Veksler

Researcher - Peter Kiprop

Introduction

My summer research project examined whether a structured prompt can improve the quality of a large language model's output. I tested the CRAFT framework - Context, Role, Action, Format, and Tone - against an information-matched but less structured baseline. The goal was not to compare a vague prompt with a detailed one. Both prompt conditions provided the model with the same task facts and input; CRAFT changed how the instructions were organized, framed, and presented.

The project used a 50-task benchmark across Coding and Debugging, Data Analysis, Finance and Business, Policy and Ethics, Instructional Design, and Professional Communication. Each task ran on two large language models under both baseline and CRAFT conditions, creating 100 paired task-by-model comparisons. I scored outputs using a blinded 10-point rubric for constraint adherence, logical accuracy, and completeness.

At the beginning of the summer, I had hoped to learn how to evaluate a prompt engineering framework rigorously rather than relying on individual examples or subjective impressions. My professional goals were to strengthen my understanding of artificial intelligence, research design, data analysis, and the practical limitations of prompt engineering. I also wanted to learn how to make an AI experiment reproducible: how to construct a task set, define measurable outcomes, evaluate model responses fairly, and interpret results even when they do not support the original hypothesis.

Main Body

Research methodology and task construction

I designed the benchmark so that each domain tested a different type of reasoning or communication. I adapted the Coding and Debugging tasks from OpenAI's HumanEval benchmark, which assesses the functional correctness of programs synthesized from docstrings [1]. I then reframed selected problems as applied scenarios and introduced original bugs to check against expected behavior. The finance and Business tasks used comparable computational structures but placed them in financial contexts, making the answers numerically verifiable while giving me space to compare similar reasoning patterns across domains.

The Data Analysis tasks drew on documented statistical and decision-making phenomena, including Simpson's paradox, survivorship bias, regression to the mean, and base-rate neglect[2,3]. The numbers in the tasks were original, but the reasoning patterns were based on well-established concepts. For example, one task used Simpson's paradox to test whether a model could recognize that an aggregate result may reverse the trend observed within individual groups.

For the Policy and Ethics, Education, and Professional Communication domains, a single numerical answer could not determine whether the response was good. The central risk was that an evaluator might reward a response merely for being long, polished, or highly structured. To reduce that risk, I constructed five observable constraints for each task. For example, instead of requiring a response to 'show awareness of stakeholders,' a rubric could require it to "name at least three distinct stakeholder groups affected by the decision." This made the evaluation more transparent and less dependent on an evaluator's personal view of what sounds thoughtful.

I also considered using QRData, a published ACL 2024 benchmark that evaluates data-based statistical and causal reasoning with real-world data [4]. QRData was useful as related work, but it did not fit this study’s instrument. Its questions commonly rely on separate data sheets, while this study required self-contained text inputs that could be presented identically rather than extended responses, meaning they could not meaningfully test the CRAFT format and Tone dimensions. Constructing smaller, self-contained data analysis was hence a methodological decision, not just a convenient shortcut.

A major lesson from task construction was that sources must be verified, not copied. While adapting one HumanEval problem, I found that the published docstring gave the wrong median for a list of values. I also caught an early finance scenario whose cash-flow figures did not actually create the floor breach; instead described in its rubric. Both cases showed me that a benchmark error can silently produce false results by penalizing correct answers. This experience directly advanced my objectives of building valid, testable, and research instruments.

Reflection on the field

This project gave me a more nuanced view of prompt engineering. I initially viewed CRAFT as a possible universal framework for producing better responses, but by the end of the project, I understood it more accurately as a prompt-authoring checklist: a way to make instructions visible, organized, and reproducible. Although the five components are stored separately, the design cannot determine whether Role, Format, Tone, or any other component caused a result independently.

This limitation is consistent with Anthropic’s evaluation-first guidance. They recommend defining success criteria, testing against those criteria, and recognizing that not every failure

should be solved by adding more prompt engineering [5]. Its context engineering guidance similarly emphasizes using the smallest high-signal set of information that fully supports the expected behavior, then adding instructions or examples in response to observed failure modes [6]. My results are consistent with that view.

I also learned that CRAFT could be useful with an agentic system. CRAFT by itself is not a complete agentic skill. A reliable skill also needs task-specific context, retrieval or tools, verification, memory, and a process for recovering from errors. Context engineering is broader than prompt engineering because it covers the full set of information available to a model, including system instructions, tools, external data, and conversation history [6]. In that larger architecture, CRAFT can function as a readable specification for stable instructions. For example, a financial analysis skill could store an analyst-perspective role, a repeatable research sequence as Actions, and a memo contract as format, while receiving company data and retrieved documents as task-specific context.

This changed my view of my career interests. It clarified that writing a one-time prompt is only a small part of building a useful AI system. As models become more capable, the work shifts toward choosing the right context, designing workflows, verifying outputs, building tools, and evaluating reliability.

This is my first formal research project, and the experience met my learning expectations even though it did not confirm my original hypothesis. I expected to learn whether CRAFT worked; I learned something more useful about how to test whether it works. The unexpected result forced me to focus on validity, ceiling effects, model drift, evaluator disagreement, and incomplete evidence. If the project had produced only a positive result, I might have concluded too quickly that a longer and more structured prompt was always better. Instead, I learned that a

strong baseline and an honest interpretation are more valuable than a result that merely confirms an expectation.

Conclusion

This summer research experience positively shaped how I think about artificial intelligence and research itself. It strengthened my technical understanding of prompt design, benchmarking, data analysis, model evaluation, reproducible workflows, and, more importantly, it taught me to treat an unexpected result as information rather than failure. It did not convince me that prompt engineering alone should be my career goal. Instead, it showed me that a more durable interest is the broader design of reliable AI systems: systems that manage context carefully, use tools appropriately, test outputs, preserve evidence, and communicate uncertainty. These skills are relevant to future work in artificial intelligence, finance, business, and technology because they focus on making AI systems useful and trustworthy, not merely fluent.

Although the main result did not show the improvement I expected from CRAFT, the project still benefited my future goals. It gave me experience with a real research problem whose answer was not known in advance. It also showed me that technical progress does not eliminate the need for human judgment; it changes where that judgment matters most. Going forward, I will approach AI claims with more curiosity, stronger evidence standards, and a clearer interest in the systems around a model, not only the prompt inside it.

References

- [1]hen, M., Tworek, J., Jun, H., Yuan, Q., Pinto, H. P. de O., Kaplan, J., Edwards, H., Burda, Y., Joseph, N., Brockman, G., Ray, A., Puri, R., Krueger, G., Petrov, M., Khlaaf, H., Sastry, G., Mishkin, P., Chan, B., Gray, S., ... Zaremba, W. (2021). Evaluating large language models trained on code. *arXiv:2107.03374 [Cs]*. <https://arxiv.org/abs/2107.03374>
- [2]Bickel, P. J., Hammel, E. A., & O'Connell, J. W. (1975). Sex bias in graduate admissions: Data from Berkeley. *Science*, 187(4175), 398–404.
<https://doi.org/10.1126/science.187.4175.398>
- [3]Tversky, A., & Kahneman, D. (1974). Judgment under uncertainty: Heuristics and Biases. *Science*, 185(4157), 1124–1131.
<https://www.science.org/doi/10.1126/science.185.4157.1124>
- [4]Liu, X., Wu, Z., Wu, X., Lu, P., Chang, K.-W., & Feng, Y. (2024). Are {LLM}s capable of data-based statistical and causal reasoning? Benchmarking advanced quantitative reasoning with data. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Findings of the Association for Computational Linguistics: ACL 2024* (pp. 9215--9235). Association for Computational Linguistics. <https://doi.org/10.18653/v1/2024.findings-acl.548>
- [5]Anthropic. (2025b). *Prompt engineering overview*. Anthropic. Claude API Docs.
<https://platform.claude.com/docs/en/build-with-claude/prompt-engineering/overview>
- [6]Anthropic. (2025a). *Effective context engineering for AI agents*. Anthropic. Anthropic.Com.
<https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents>